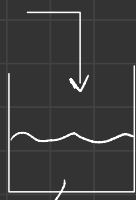


Miarowanie - dodawanie kroplami titranta do analitu
↑
substancja, której stężenie dokładnie znamy

Miarowanie alkaliometryczne (kwas + zasada)



dodajemy kroplami roztwór NaOH $c = 1 \frac{\text{mol}}{\text{dm}^3}$ *tytuł*



roztwór HCl

$$c = 1 \frac{\text{mol}}{\text{dm}^3}$$
$$V = 100 \text{ cm}^3$$

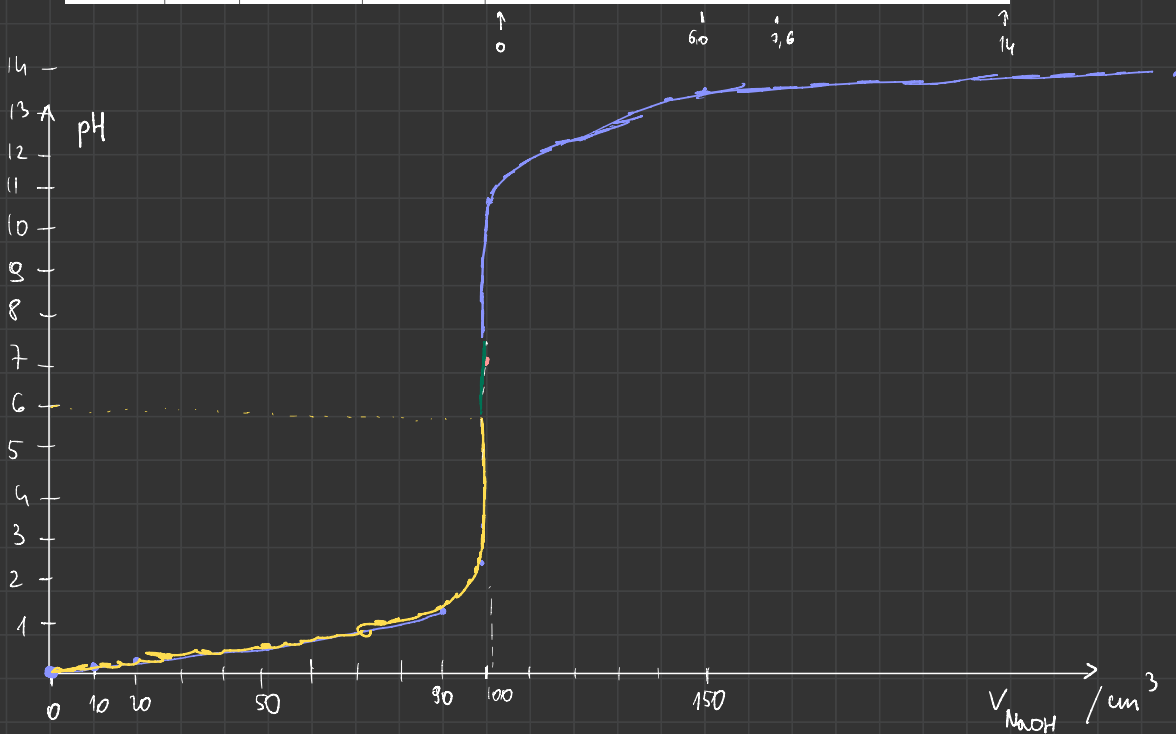
analit

$$\text{pH} = -\log_{10}[\text{H}^+] = -\log_{10} 1 = 0$$

↓
HCl

$$\underline{10^x = 1}$$

blekit bromotymolowy	6,0 – 7,6	żółta	niebieska	
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$$\text{HCl: } c = 1 \text{ mol/dm}^3 \quad V = 100 \text{ cm}^3$$

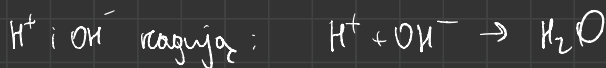
$$n_{\text{H}^+} = 1 \cdot 0,1 = 0,1$$

$$c = \frac{n}{V}$$

$$\text{NaOH: } c = 1 \text{ mol/dm}^3 \quad V = 10 \text{ cm}^3$$

$$n_{\text{OH}^-} = 1 \cdot 0,01 = 0,01$$

$$n = c \cdot V$$



wgłc zostaje $n_{\text{H}^+} = 0,09 \text{ mol}$

	n_0	Δn	
H^+	0,1	-0,01	0,09
OH^-	0,01	-0,01	0

$$V = 100 \text{ cm}^3 + 10 \text{ cm}^3 = 110 \text{ cm}^3 = 0,11 \text{ dm}^3$$

$$C_{\text{H}^+} = \frac{n}{V} = \frac{0,09 \text{ mol}}{0,11 \text{ dm}^3} = 0,82 \text{ mol/dm}^3$$

$$\text{pH} = -\log_{10} 0,82 = 0,09$$

V_{NaOH}	0 dm^3	10 cm^3	20 cm^3	50 cm^3
$V_r = (V_{\text{NaOH}} + V_{\text{HCl}})$ 100 cm^3 1 mol/dm^3	100 cm^3	110 cm^3	120 cm^3	150 cm^3
$n_{\text{OH}^-} \rightarrow c_{\text{OH}^-} \cdot V_{\text{NaOH}}$	0 mol	$0,01 \text{ mol}$	$0,02 \text{ mol}$	$0,05 \text{ mol}$
$n_{\text{H}^+}^0 = 0,1 \text{ mol}$	$0,1 \text{ mol}$	$0,1 \text{ mol}$	$0,1 \text{ mol}$	$0,1 \text{ mol}$
$n_{\text{H}^+}(\text{po reakcji})$	$0,1 \text{ mol}$	$0,09 \text{ mol}$	$0,08 \text{ mol}$	$0,05 \text{ mol}$
c_{H^+}	1 mol/dm^3	$0,82 \text{ mol/dm}^3$	$0,67 \text{ mol/dm}^3$	$0,33 \text{ mol/dm}^3$
pH	0	0,03	0,18	0,48

$$c = \frac{0,08 \text{ mol}}{0,12 \text{ dm}^3} = 0,67 \text{ mol/dm}^3 \quad \text{pH} = -\log_{10} 0,67 = 0,18$$

$$c = \frac{0,05 \text{ mol}}{0,15 \text{ dm}^3} = 0,33 \text{ mol/dm}^3 \quad \text{pH} = -\log_{10} 0,33 = 0,48$$

V_{NaOH}	90 cm^3	99 cm^3	$99,9 \text{ cm}^3$	100 cm^3
V_r	190 cm^3	199 cm^3	$199,9 \text{ cm}^3$	200 cm^3
n_{OH^-}	$0,09 \text{ mol}$	$0,099 \text{ mol}$	$0,0999 \text{ mol}$	$0,1 \text{ mol}$
$n_{\text{H}^+}^0$	$0,1 \text{ mol}$	$0,1 \text{ mol}$	$0,1 \text{ mol}$	$0,1 \text{ mol}$
$n_{\text{H}^+}(\text{po r.})$	$0,01 \text{ mol}$	$0,001 \text{ mol}$	$0,0001 \text{ mol}$	0 mol
c_{H^+}	$0,053 \text{ mol/dm}^3$	$0,005 \text{ mol/dm}^3$	$0,0005 \text{ mol/dm}^3$	0 (10^{-7})
pH	1,28	2,3	3,3	7



$$K_w = 10^{-14} \text{ (iloczyn jonowy wody)}$$

$$[\text{H}^+] \cdot [\text{OH}^-] = 10^{-14} \text{ jeśli wypojskniwie, to}$$

$$x \cdot x = 10^{-14} \quad [\text{H}^+] = [\text{OH}^-] = x$$

$$x^2 = 10^{-14}$$

$$x = 10^{-7} = [\text{H}^+]$$

$$\text{pH} = -\log_{10} 10^{-7} = 7$$

obojętnie pH to 7

99,9 cm ³	100 cm ³	V _{NaOH}	100,1 cm ³
199,9 cm ³	200 cm ³	V _r	200,1 cm ³
0,0999 mol	0,1 mol	n _{OH-}	0,1001 mol
0,1 mol	0,1 mol	n _{H+} ^o	0,1 mol
0,0001 mol	0 mol	n _{OH-} (po r.)	0,0001 mol
0,0005 mol/dm ³	0 (10 ⁻⁷)	c _{OH-}	= 0,0005 mol/dm ³ ⇒
3,3	7	pOH	= 3,3 ⇒ 10,7

$$c = \frac{0,0001 \text{ mol}}{0,2001 \text{ dm}^3} = 0,0005$$

Jeśli zmierzyć $[OH^-]$ na $[H^+]$ 2
pOH na pH 0

Konstanta z iloczynu jonowego wody:

$$[H^+] \cdot [OH^-] = K_w$$

$$\text{w } T = 25^\circ C \\ K_w = 10^{-14}$$

$$[H^+] = \frac{K_w}{[OH^-]}$$

$$\star [H^+] = \frac{10^{-14}}{0,0005} = 2 \cdot 10^{-11}$$

$$\star pH = -\log(2 \cdot 10^{-11}) = 10,7$$

Jeśli chcemy tylko pH:

$$[H^+] \cdot [OH^-] = 10^{-14} \quad | \log_{10}$$

$$\log_{10}([H^+] \cdot [OH^-]) = \log_{10} 10^{-14}$$

$$\log_{10} [H^+] + \log_{10} [OH^-] = -14 \quad | \cdot (-1)$$

$$\underbrace{-\log_{10} [H^+]}_{pH} - \underbrace{\log_{10} [OH^-]}_{pOH} = 14$$

$$\star \boxed{pH + pOH = 14}$$

$$\text{jeśli } pOH = 3,3 \text{ to } pH = 10,7$$

$$V \quad 150 \text{ cm}^3$$

$$1000 \text{ cm}^3$$

$$V_r \quad 250 \text{ cm}^3$$

$$1100 \text{ cm}^3$$

$$n_{\text{OH}^-} \quad 0,15 \text{ mol}$$

$$1 \text{ mol}$$

$$n_{\text{H}^+} \quad 0,1 \text{ mol}$$

$$0,1 \text{ mol}$$

$$n_{\text{OH}^-} (\text{po r.}) \quad 0,05 \text{ mol}$$

$$0,9 \text{ mol}$$

$$c_{\text{OH}^-} = 0,05 / 0,25 = 0,2 \text{ mol/dm}^3$$

$$c_{\text{OH}^-} = 0,9 / 1,1 = 0,82 \text{ mol/dm}^3$$

$$\text{pOH} \quad -\log 0,2 = 0,7 \quad \text{pH} = 13,3$$

$$\text{pOH} = 0,09 \quad \text{pH} = 13,91$$

